UMER SIDDIQUI

Backend & AI Systems Engineer

Karachi, Pakistan · um3rsiddiqui99@gmail.com · +92-341-8094081

github.com/UmerSidiqui · linkedin.com/in/um3r-siddiqui · umersidiqui.github.io/Portfolio

PROFESSIONAL SUMMARY

Backend and AI systems engineer who designs and deploys production web APIs, telephony voice agents, and retrieval pipelines using FastAPI, NestJS, and PostgreSQL. Built voice receptionist workflows and dental practice-management integrations handling live calls with sub-second response times and concurrency-safe appointment booking. Experienced across system design, cloud CI/CD, and CPU-only, in-process ONNX model inference.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Back-End: FastAPI, NestJS, Node.js, REST APIs, WebSockets

AI & LLM Orchestration: LangGraph, LangChain, Model Context Protocol (MCP), Claude, OpenAI API, ElevenLabs, RAG, ONNX Runtime

Databases & Caching: PostgreSQL, pgvector, Redis, Firestore, SQLite

Cloud & DevOps: AWS (ECS, RDS, S3), GCP, Docker, GitHub Actions CI/CD

Computer Vision & ML: PyTorch, YOLO, ResNet, OpenCV

PROFESSIONAL EXPERIENCE

Saynt AI — Senior AI Engineer **Jun 2025 – Present · Remote**

Healthcare Voice AI & Practice Management Platform

Platform Engineering *(NestJS · TypeScript · PostgreSQL · AWS)*

- Built and maintained NestJS backend services for authentication, appointment scheduling, patient records, call logging, and notifications across multi-location dental clinics.
- Implemented authentication using Google OAuth 2.0 and JWT token verification with tenant isolation across services.
- Designed relational PostgreSQL schemas with TypeORM to track patient appointments, provider schedules, and audit records.
- Built Docker containers and automated CI/CD deployment pipelines using GitHub Actions to deploy backend services to AWS.

Conversational Voice Engine & Telephony Integration *(FastAPI · Twilio · ElevenLabs · ONNX)*

- Built a real-time telephony webhook service in FastAPI integrating Twilio voice streams with ElevenLabs conversational voice agents to handle inbound patient phone calls.
- Implemented bidirectional webhook tool-calling connecting directly to dental practice management systems (PMS) for live calendar slot lookups, provider availability checks, and appointment creation.
- Designed a concurrency-safe booking lock mechanism to prevent double-booking during simultaneous inbound calls.
- Benchmarked end-to-end voice latency metrics on live calls: ASR processing (20–80 ms), streaming TTS generation (65–90 ms), and practice management tool execution (<1.7 s), maintaining low-latency conversational turn-taking.
- Integrated CPU-only, in-process ONNX inference using quantized embeddings (all-MiniLM-L6-v2) for sub-100ms semantic RAG over clinic policies and insurance FAQs without third-party API latency or cost.
- Implemented automated PII log redaction for patient data privacy, call transcript ingestion, and asynchronous SMS appointment confirmations.

Erly Stage Studios — Jr. AI Developer **Oct 2024 – Jun 2025 · Remote**

Conversational AI & Retrieval Systems

WhatsApp Meeting Scheduling Agent *(FastAPI · LangGraph · Model Context Protocol)*

- Built an autonomous conversational scheduling assistant integrating WhatsApp webhooks with a multi-server Model Context Protocol (MCP) architecture.
- Developed dedicated MCP tools for Google Calendar event creation, conflict checks, and attendee invites using OAuth2 token management.
- Created a temporal reasoning MCP server to resolve natural-language relative expressions (e.g., 'next Tuesday at 3pm') into standardized ISO timestamps with dynamic timezone conversions.
- Structured multi-turn state graphs using LangGraph and FastAPI, collaborating with an Anthropic engineer on prompt design, session state persistence, and error recovery flows.

Product Q&A Hybrid Retrieval Engine *(Python · Flask · PostgreSQL · pgvector)*

- Built an e-commerce catalog search service combining exact relational SQL filtering with pgvector similarity search over product embeddings.
- Implemented conversational query rephrasing and chat history persistence to maintain multi-turn context across product inquiries.

NCL, NED University of Engineering & Technology — AI & Telemetry Intern **Apr 2024 – Oct 2024 · Karachi**

- Built Python data processing and EDA pipelines (Pandas) to analyze high-frequency automotive CAN-bus trip sensor logs across vehicle speed, engine RPM, fuel trims, and coolant temperatures.

ADDITIONAL EXPERIENCE

Supporting Engineer, HVAC Energy & Performance Audit **6-Day Onsite Audit, Oct 2026 · Karachi**

Commercial Facility (British Council Library)

- Installed 3-phase power quality analyzers and data loggers across commercial VRV outdoor units during a 6-day field audit; wrote Python scripts for exploratory data analysis (EDA) to synchronize electrical power draw with thermodynamic compressor load cycles and identify operating imbalances.
- Analyzed electrical power logs alongside indoor cooling/return temperatures, suction/discharge pressures, acoustic/vibration measurements, thermal imaging, and room airflow data; identified high refrigerant leak frequency and equipment degradation on a 10-year-old system, delivering data-backed recommendations for full system replacement.

INDEPENDENT PROJECTS

Multi-Object Visual Reasoning & Orientation Pipeline *(PyTorch · YOLO · ResNet-18 · ONNX Runtime · OpenCV · NetworkX)*

- Built an automated computer vision pipeline to solve complex multi-symbol visual matching and orientation challenges under geometric clutter.
- Built a custom web-based annotation tool (JavaScript) to label symbol datasets locally; fine-tuned YOLO and ResNet models on Google Colab (NVIDIA T4 GPU) and exported lightweight, CPU-optimized ONNX models to run efficiently on low-spec hardware.
- Fine-tuned a nano YOLO model on custom-labeled images to detect and isolate prompt icons and candidate scene bounding boxes under dynamic confidence thresholds.
- Preprocessed candidate crops into 8-way rotational views (0°–315° in 45° increments) and evaluated them using a fine-tuned ResNet-18 Siamese network running in ONNX Runtime; computed pairwise cosine similarities across rotational tensors.
- Formulated the symbol-matching assignment as a bipartite graph in NetworkX, running maximum weight bipartite matching to determine the ordered sequence of target coordinates with an automated validation threshold (≥ 0.60).

Video Surveillance Violence Detection System *(Python · YOLO · OpenCV)*

- Fine-tuned a YOLO object detection model on curated video surveillance datasets labeled for physical altercations versus non-violent activities.
- Built automated frame extraction and confidence-thresholded event detection logic in OpenCV for real-time video security monitoring.

RESEARCH

Heuristics-Driven Soft Actor-Critic (SAC) Reinforcement Learning *(Deep RL · PyTorch · OpenAI Gym)*

- Reproduced and extended the base paper: Beikmohammadi & Magnússon, 'Accelerating actor-critic-based algorithms via pseudo-labels derived from prior knowledge', Information Sciences 661 (2024).
- Formulated a novel critic-side heuristic modification to the value function update in Soft Actor-Critic (SAC), supplementing the base paper's actor-side pseudo-labels.
- Evaluated across continuous control benchmark environments (CartPoleContinuous-v1, MountainCarContinuous-v0, LunarLanderContinuous-v2, Pendulum-v1, BipedalWalker-v3) against Stable-Baselines3 SAC, TD3, DDPG, PPO, and the baseline Accelerated SAC over multiple random seeds, tracking cumulative return trajectories and sample efficiency.
- Conducted research under the supervision of Dr. Ali Baig (Assistant Professor, NED University) alongside a departmental co-researcher; manuscript submitted and currently under peer review.

EDUCATION

NED University of Engineering & Technology, Karachi **2022 – 2026**

Bachelor of Engineering, Electrical Engineering · CGPA: 2.60 / 4.00 · Final Year Design Project: Heuristics-Driven Actor-Critic RL

Bahria College Karsaz, Karachi **2020 – 2022**

Higher Secondary (Intermediate), Pre-Engineering · BIEK Board · 996 / 1100 marks

CAA Model School and College, Karachi **2018 – 2020**

Secondary School (Matriculation), Science · BSEK Board · 755 / 850 marks

LANGUAGES

English (Professional), Urdu (Native), Turkish (Beginner)